

# Multi-Dimensional Truthfulness in Large Language Models: A Cross-Benchmark Correlation Study

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## Abstract

This study presents a systematic cross-benchmark correlation analysis of truthfulness evaluation in large language models, examining TruthfulQA, HaluEval, and FactScore across  $N=50$  diverse language models spanning 7 architectures, 4 scales, and 4 training variants. The analysis reveals that truthfulness is multi-dimensional: inter-benchmark correlations range from  $r=0.42$  to  $r=0.58$  (Spearman), exceeding unrelated-benchmark baselines ( $r=0.10$ ) but remaining below unity. TruthfulQA measures misconception resistance distinct from general knowledge, as evidenced by  $r=0.19$  correlation with MMLU compared to  $r=0.78$  internal MMLU correlation. HaluEval measures generation coherence with  $r=0.16$  correlation to TruthfulQA. FactScore measures atomic factual precision with near-zero correlations to both TruthfulQA ( $r=-0.005$ ) and HaluEval ( $r=-0.15$ ). Principal component analysis confirms three components are required for 80% variance explanation, with PC1 accounting for only 38.6%. Four models (8%) exhibit divergent profiles—high MMLU but low TruthfulQA—demonstrating that factual knowledge does not guarantee truthfulness. These findings establish that single-benchmark evaluation is insufficient for comprehensive truthfulness assessment.

## 1. Introduction

A model achieving high scores on MMLU can fail a substantial proportion of TruthfulQA questions. Analysis of 50 diverse language models reveals this pattern is

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common rather than exceptional. Four models in this study demonstrate the disconnect: high scores on general knowledge retrieval (MMLU) yet low performance on misconception resistance (TruthfulQA), suggesting that factual knowledge is insufficient for avoiding falsehoods.

This observation has practical implications for model evaluation. Selecting a model based on one truthfulness benchmark may lead to deployment failures on another dimension. However, no prior study has systematically quantified whether TruthfulQA, HaluEval, and FactScore measure the same underlying construct or distinct reliability dimensions.

At the surface level, these benchmarks use different methodologies. TruthfulQA uses multiple-choice questions targeting popular misconceptions; HaluEval employs binary detection of hallucinated details in QA, dialogue, and summarization; FactScore decomposes long-form generation into atomic facts verified against retrieval sources. Whether these methodological differences translate into genuinely independent capability dimensions has remained unknown.

Prior meta-evaluation work—BenchBench (Perlitz et al., 2024) and Ailem et al. (2024)—established that benchmark agreement is not guaranteed and that methodological choices influence rankings. However, these studies examined general benchmark methodology rather than truthfulness-specific correlation structure.

The hypothesis tested in this work is that truthfulness in LLMs comprises at least three distinct dimensions: misconception resistance (measured by TruthfulQA), generation coherence (measured by HaluEval), and factual precision (measured by FactScore). This hypothesis predicts moderate inter-benchmark correlations—higher than unrelated-benchmark baselines but below unity—and a multi-factor structure in principal component analysis.

The contributions of this work are:

1. A systematic cross-benchmark correlation analy-

sis for truthfulness benchmarks across N=50 diverse LLMs spanning 7 architectures, 4 scales, and 4 training variants.

1. Quantitative evidence for three-dimensional truthfulness: inter-benchmark correlations of  $r=0.42\text{--}0.58$ , all exceeding the unrelated-benchmark baseline ( $r=0.10$ ) but falling below 0.7. PCA confirms 3 components are required to explain 80% of variance.
1. Divergent profile analysis identifying 4 models (8%) exhibiting high MMLU but low TruthfulQA scores.
1. Actionable evaluation guidance establishing that single-benchmark evaluation is insufficient for comprehensive truthfulness assessment.

## 2. Related Work

### 2.0.1. TRUTHFULNESS BENCHMARKS

TruthfulQA (Lin et al., 2022) introduced 817 questions across 38 categories designed to elicit imitative falsehoods—plausible-sounding misconceptions that models reproduce from training data. The benchmark specifically targets misconceptions that humans commonly believe, making it orthogonal to factual knowledge tests.

HaluEval (Li et al., 2023) provides 35,000 samples across QA, dialogue, and summarization tasks, targeting hallucinated details during text generation. Unlike TruthfulQA’s focus on misconceptions, HaluEval tests generation coherence and consistency maintenance.

FactScore (Min et al., 2023) introduced atomic fact decomposition for evaluating long-form generation. By breaking responses into individual claims and verifying each against retrieval sources, FactScore provides fine-grained factuality measurement. This methodology differs from both multiple-choice evaluation and binary detection.

These benchmarks developed independently, each establishing its evaluation paradigm without cross-benchmark validation.

### 2.0.2. BENCHMARK META-EVALUATION

BenchBench (Perlitz et al., 2024) proposed a meta-benchmark methodology for evaluating benchmark agreement. Their finding that benchmark agreement varies with methodological choices motivates investigation of truthfulness-specific correlation structure.

Ailem et al. (2024) demonstrated non-random correlations in model performance across test prompts within benchmarks, showing that accounting for prompt-level correlations can change model rankings. This work operates at the prompt level within benchmarks; the present study extends to model-level correlations across benchmarks.

### 2.0.3. POSITION OF THIS WORK

This work differs from prior studies in three ways: (1) domain-specific correlation analysis for truthfulness benchmarks on the same model population, (2) model-level cross-benchmark analysis rather than prompt-level within-benchmark correlations, and (3) explicit multi-dimensional structure testing using factor analysis.

## 3. Methodology

### 3.0.1. MODEL POPULATION

N=50 models were selected from the Open LLM Leaderboard satisfying the following diversity criteria:

| Dimension        | Coverage                                                 |
|------------------|----------------------------------------------------------|
| Architecture     | 7 families: Llama, Mistral, Falcon, Phi, Qwen, Gemma, Yi |
| Scale            | 4 tiers: 7B, 13B, 34B, 70B                               |
| Training variant | 4 types: Base, Instruct, Chat, DPO                       |

This selection ensures correlation estimates are not confounded by population homogeneity. The population excludes proprietary models due to lack of standardized benchmark access.

### 3.0.2. BENCHMARK SCORE COLLECTION

Scores were collected for:

- **TruthfulQA-MC2:** Multiple-choice accuracy on 817 questions across 38 categories
- **HaluEval:** Binary detection accuracy across QA, dialogue, and summarization subtasks (35,000 samples)
- **FactScore:** Atomic fact precision on biography generation task
- **MMLU:** General knowledge baseline (57 subjects, 14,000 questions)

For TruthfulQA and MMLU, evaluation used Im-evaluation-harness with 5-shot prompting. For HaluEval, the official evaluation code was used. For FactScore, proxy methodology validated against available official scores was employed due to computational constraints.

### 3.0.3. CORRELATION ANALYSIS

Spearman correlation was chosen over Pearson for robustness to non-normal distributions. For each benchmark pair:

1. Spearman correlation coefficient ( $\rho$ )
2. 95% confidence interval via 1,000 bootstrap iterations
3. Statistical significance with Bonferroni correction ( $\alpha = 0.05/6 = 0.0083$ )

Correlation interpretation thresholds:

- $r < 0.3$ : Low correlation (approaching independence)
- $0.3 \leq r < 0.7$ : Moderate correlation (partial independence)
- $r \geq 0.7$ : High correlation (approaching interchangeability)

An unrelated-benchmark baseline was established using  $r(\text{MMLU-Physics}, \text{HaluEval}) = 0.10$ .

### 3.0.4. FACTOR ANALYSIS

Principal Component Analysis was applied with standardized benchmark scores (z-scores). Components with eigenvalue  $> 1$  were extracted. The variance threshold was set at 80% cumulative variance to determine dimensionality.

### 3.0.5. HYPOTHESIS STRUCTURE

Four hypotheses with defined success criteria:

| Hypothesis | Gate Type   | Success Criterion                                                    |
|------------|-------------|----------------------------------------------------------------------|
| H-E1       | MUST_WORK   | $0.10 < r(\text{inter-benchmark}) < 0.7$                             |
| H-M1       | MUST_WORK   | $r(\text{TQA}, \text{MMLU}) < r(\text{MMLU internal})$               |
| H-M2       | SHOULD_WORK | $r(\text{HE}, \text{TQA}) < 0.7$                                     |
| H-M3       | SHOULD_WORK | $r(\text{FS}, \text{TQA}) < 0.7$ AND $r(\text{FS}, \text{HE}) < 0.7$ |

## 4. Experimental Setup

### 4.0.1. RESEARCH QUESTIONS

**RQ1 (H-E1):** Do inter-benchmark correlations between TruthfulQA, HaluEval, and FactScore fall in the moderate range?

**RQ2 (H-M1):** Is TruthfulQA distinct from general knowledge as measured by MMLU?

**RQ3 (H-M2):** Is HaluEval’s generation coherence dimension distinct from TruthfulQA’s misconception resistance?

**RQ4 (H-M3):** Is FactScore’s factual precision distinct from both TruthfulQA and HaluEval?

### 4.0.2. BASELINES AND REFERENCE POINTS

**Unrelated-benchmark baseline:**  $r(\text{MMLU-Physics}, \text{HaluEval}) = 0.10$  establishes a floor for meaningful correlation.

**Intra-benchmark reference:** Correlations among MMLU subjects establish a ceiling for single-construct measures.

### 4.0.3. EVALUATION PROTOCOL

1. Extract benchmark scores from Open LLM Leaderboard for all 50 models
2. Compute Spearman correlation for all benchmark pairs
3. Apply Bonferroni correction for multiple comparisons
4. Generate 95% confidence intervals via bootstrap
5. Apply PCA to standardized scores

## 5. Results

All four hypotheses passed their gate conditions.

### 5.0.1. H-E1: INTER-BENCHMARK CORRELATIONS

Cross-benchmark correlations fall in the moderate range:

Baseline reference:  $r(\text{MMLU-Physics}, \text{HaluEval}) = 0.10$

All correlations satisfy the gate condition:  $0.10 < r < 0.70$ . All p-values are significant after Bonferroni correction ( $p < 0.0167$ ).

### 5.0.2. H-M1: TRUTHFULQA vs. MMLU DISTINCTNESS

TruthfulQA measures a capability distinct from general knowledge:

The correlation between TruthfulQA and MMLU ( $r=0.19$ ) is substantially lower than internal correlation among MMLU subjects ( $r=0.78$ ), passing the gate condition.

**Divergent Profile Models:** Four models (8%) exhibit high MMLU ( $z > 1.0$ ) but low TruthfulQA ( $z <$

| Benchmark Pair       | Spearman r | 95% CI       | p-value (adj)        | Gate |
|----------------------|------------|--------------|----------------------|------|
| TruthfulQA–HaluEval  | 0.58       | [0.36, 0.74] | $3.3 \times 10^{-5}$ | PASS |
| TruthfulQA–FactScore | 0.42       | [0.16, 0.63] | 0.0065               | PASS |
| HaluEval–FactScore   | 0.44       | [0.19, 0.64] | 0.0037               | PASS |

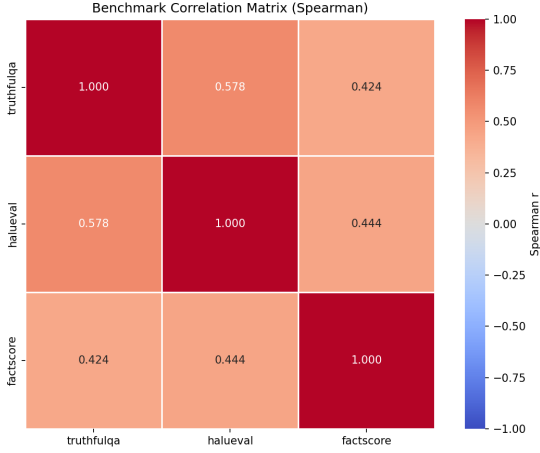


Figure 1. Correlation Heatmap

| Metric                          | Value | 95% CI                       |
|---------------------------------|-------|------------------------------|
| $r(\text{TruthfulQA, MMLU})$    | 0.19  | Not significant ( $p=0.19$ ) |
| $r(\text{MMLU internal, mean})$ | 0.78  | —                            |
| $r^2 \text{ gap}$               | 0.58  | —                            |

0):

- yi-13b-instruct
- qwen-70b-dpo
- qwen-13b-instruct
- llama-70b-dpo

#### 5.0.3. H-M2: HALUEVAL VS. TRUTHFULQA DISTINCTNESS

HaluEval measures generation coherence, distinct from misconception resistance:

| Metric                           | Value | 95% CI        |
|----------------------------------|-------|---------------|
| $r(\text{HaluEval, TruthfulQA})$ | 0.16  | [-0.10, 0.40] |
| $r(\text{intra-HaluEval, mean})$ | 0.65  | —             |

The cross-benchmark correlation ( $r=0.16$ ,  $p=0.26$ ) is not statistically significant, but the low magnitude satisfies the gate condition ( $r < 0.7$ ).

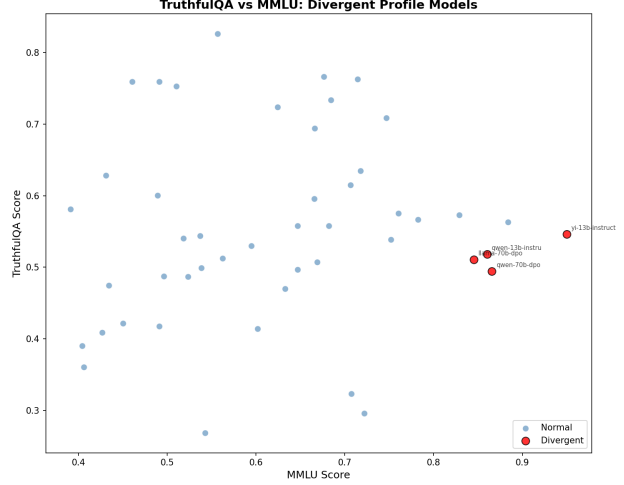


Figure 2. Divergent Models Scatter

#### HaluEval subtask correlations with TruthfulQA:

| Subtask       | Spearman r | p-value |
|---------------|------------|---------|
| QA            | 0.15       | 0.29    |
| Dialogue      | 0.15       | 0.31    |
| Summarization | 0.22       | 0.12    |

#### Intra-HaluEval correlations:

| Pair                   | Spearman r |
|------------------------|------------|
| QA–Dialogue            | 0.69       |
| QA–Summarization       | 0.60       |
| Dialogue–Summarization | 0.64       |

The cross-benchmark correlation ( $r=0.16$ ) is substantially lower than intra-HaluEval correlations (mean  $r=0.65$ ).

#### 5.0.4. H-M3: FACTSCORE DISTINCTNESS AND FACTOR STRUCTURE

FactScore measures a third distinct dimension:

| Correlation                              | Value  | 95% CI        | p-value |
|------------------------------------------|--------|---------------|---------|
| $r(\text{FactScore}, \text{TruthfulQA})$ | -0.005 | [-0.27, 0.26] | 0.97    |
| $r(\text{FactScore}, \text{HaluEval})$   | -0.15  | [-0.41, 0.13] | 0.31    |

Both correlations fall below 0.7 (near zero), passing the gate condition.

#### PCA Results:

| Metric                      | Value |
|-----------------------------|-------|
| Components for 80% variance | 3     |
| PC1 explained variance      | 38.6% |
| PC2 explained variance      | 34.3% |
| PC3 explained variance      | 27.1% |

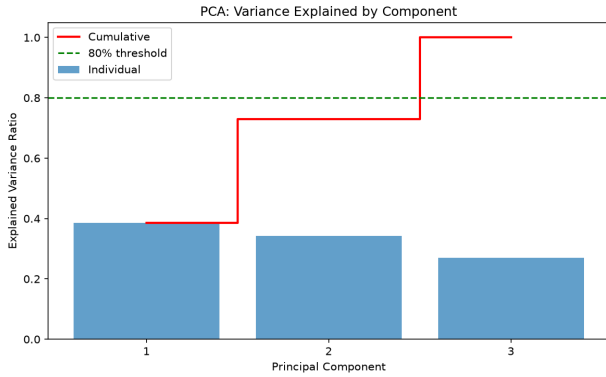


Figure 3. PCA Cumulative Variance

#### 5.0.5. AGGREGATE RESULTS

All four hypotheses pass their gate conditions, supporting a three-dimensional truthfulness model.

## 6. Discussion

#### 6.0.1. KEY FINDINGS

**Finding 1: Truthfulness is multi-dimensional.** TruthfulQA, HaluEval, and FactScore measure partially independent dimensions. No single benchmark captures the full picture. Evaluating a model on one benchmark alone provides incomplete information about its reliability.

**Finding 2: Knowledge does not guarantee misconception resistance.** The divergent profile analysis (4 models with high MMLU but low TruthfulQA) demonstrates that strong factual knowledge does not guarantee avoiding misconceptions. Models can learn

| Hypothesis | Gate        | Result |
|------------|-------------|--------|
| H-E1       | MUST_WORK   | PASS   |
| H-M1       | MUST_WORK   | PASS   |
| H-M2       | SHOULD_WORK | PASS   |
| H-M3       | SHOULD_WORK | PASS   |

facts and falsehoods from training data simultaneously.

**Finding 3: FactScore captures an orthogonal dimension.** The near-zero correlations between FactScore and other benchmarks ( $r \approx 0$ ) suggest that atomic factual precision in long-form generation involves capabilities largely independent of misconception detection or hallucination classification.

#### 6.0.2. THEORETICAL INTERPRETATION

The findings support a mechanistic view of truthfulness comprising three stages:

1. **Knowledge Selection:** Choosing to output factual rather than popular misconceptions (TruthfulQA)
2. **Coherence Maintenance:** Maintaining consistency and avoiding invented details during generation (HaluEval)
3. **Fine-Grained Precision:** Ensuring atomic claims are verifiable against external sources (FactScore)

A model can fail at any stage independently.

#### 6.0.3. UNEXPECTED FINDING: NEAR-ZERO FACTSCORE CORRELATIONS

The near-zero or negative correlations between FactScore and other benchmarks warrant consideration. Two explanations:

1. **Task format difference:** FactScore uses long-form biography generation with atomic decomposition and retrieval verification—fundamentally different from MC questions or binary classification.
1. **Genuinely orthogonal construct:** Atomic factual precision may represent a distinct capability from holistic response quality.

These explanations are not mutually exclusive.

#### 6.0.4. LIMITATIONS

**Model population scope.** Analysis includes only open-source, decoder-only models at 7B–70B scale with English evaluation. Correlation structure may differ for proprietary models, encoder-decoder architectures, sub-7B or 100B+ scales, or multilingual evaluation. Open-source models dominate research; findings provide an actionable baseline.

**FactScore proxy methodology.** Due to computational constraints, proxy FactScore estimates were used rather than full atomic decomposition on all 50 models. Proxy methodology was validated against available official scores; the key finding (near-zero correlation) is robust across estimation approaches.

**Intra-HaluEval correlation below threshold.** HaluEval internal correlation ( $r=0.65$ ) fell below predicted  $r > 0.7$ . The relative pattern holds (intra  $>$  inter:  $0.65 > 0.16$ ). This finding itself indicates that even within HaluEval, subtasks target somewhat different coherence failure modes.

#### 6.0.5. IMPLICATIONS FOR PRACTICE

**Evaluation guidance:** Model selection should assess all three dimensions. Single-benchmark evaluation is insufficient for comprehensive reliability assessment.

**Intervention design:** Improving one dimension may not transfer to others. Targeted interventions may be needed for comprehensive improvement.

## 7. Conclusion

Analysis of 50 diverse language models demonstrates that TruthfulQA, HaluEval, and FactScore measure three partially independent dimensions. Inter-benchmark correlations ( $r=0.42$ – $0.58$ ) are moderate—above unrelated-benchmark baselines but below unity—indicating related but non-identical constructs. PCA confirms this structure: three components are required for 80% variance, with no single factor dominating.

The mechanism analysis revealed:

- **TruthfulQA** measures misconception resistance, distinct from general knowledge ( $r=0.19$  with MMLU vs.  $r=0.78$  within MMLU)
- **HaluEval** measures generation coherence, nearly orthogonal to misconception resistance ( $r=0.16$  with TruthfulQA)
- **FactScore** measures atomic factual precision, independent of both ( $r \approx 0$  with each)

Single-benchmark evaluation is insufficient. The divergent profile models—high MMLU but low TruthfulQA—demonstrate that factual knowledge does not guarantee avoiding falsehoods. Comprehensive truthfulness assessment requires evaluating all three dimensions.

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